CHALLENGE 16 — Disaster Relief, From Alert to Assistance

Reference: NDMA / State disaster-management portals

Background

In a disaster, information is incomplete, contradictory, and sometimes deliberately false, while communications degrade. Responders have finite boats, vehicles, and shelters. Vulnerable people need help but cannot be exposed. The platform must keep urgent requests from being lost, stop rumours without silencing real warnings, and work when the network does not.

Scenario

Severe flooding hits four districts, with roughly 40% of mobile coverage down. Reports conflict about a bridge. A fake evacuation order is circulating. Shelter capacity changes hourly. A bedridden resident needs oxygen, an elderly person without a phone is reported by a neighbour, and the same household reports through three channels. Two agencies publish conflicting road-closure data, and responders have a limited number of boats.

Your Challenge

Design a resilient disaster platform that fuses sources with explicit credibility states, targets alerts without fatigue, triages scarce resources transparently, works over low-connectivity channels, and protects vulnerable people's data with tiered privacy.

Advanced challenge dimensions

- Fuse multi-source information with credibility tiers, corroboration counts, and time decay, and display states of "verified," "corroborated," "unconfirmed," and "disputed" with provenance.
- Detect and quarantine suspected misinformation (including a fake order) for human review, and push corrections to everyone who saw the original.
- Target alerts by polygon and severity across SMS, IVR, audio, and pictographic formats, and limit repeat alerts to avoid fatigue.
- Triage help requests with a transparent vulnerability-weighted score under constrained resources and closed roads; allow responder override with full audit, and re-triage on new information.
- Deduplicate requests across households and channels (self, neighbour, responder) with an "urgent is never lost" guarantee and preserved update history.
- **Elite:** Support store-and-forward check-in over SMS- or USSD-style channels with delayed sync, conflict resolution, and "last known safe" signals.
- **Elite:** Apply privacy tiers: public views show only aggregated, thresholded counts; exact vulnerable locations are visible only to authorized responders with a purpose and expiry; include a breach simulation.

Curveballs and live stress tests

- Two agencies publish contradictory closure data for the same road. Show the dispute state and conservative routing.
- A fake evacuation order goes viral. Show detection, quarantine, correction, and measured reach of the correction.
- Shelter data goes stale. Show freshness decay and fallback instructions.

- A responder with insufficient authorization requests the vulnerable-residents list. Enforce scope, log the attempt, and show the audit entry.

Constraints and edge cases

- Do not fabricate live hazard data, evacuation routes, or shelter capacity; clearly label simulated feeds.
- Never expose vulnerable people's exact locations publicly by default.
- Provide fallback instructions when data is stale and say how old each data point is.
- Every automated priority must be explainable and overridable by an authorized human.
- Essential instructions must work offline and in at least two languages.

Expected prototype outcome

An emergency dashboard and citizen flow run against a changing flood scenario with unreliable connectivity, conflicting sources, a rumour incident, and a responder triage view with audit history.

Success metrics to report

- Share of seeded urgent requests surfaced to responders despite duplicates and channel loss (target: all).
- Time to detect and correct the injected misinformation.
- Privacy test: zero exact vulnerable-person locations in public outputs.